

Comparison of Averaging Methods for Interface Conductivities in One-Dimensional Unsaturated Flow in Layered Unsaturated Soils

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Abstract

Numerical simulations of water flow through unsaturated soils have been well studied but very few studies have addressed water movement across soil interfaces by imposing continuity of both flux and pressure head. Imposing continuity of both head and flux leads, at a local discrete level, to a nonlinear interface equation which may exhibit multiple solutions if the spatial discretization is not fine enough, for example during infiltration with sharp fronts or in the presence of very dissimilar soil layers. The non-uniqueness of solutions of the interface equation can lead to numerical errors and/or numerical oscillations. We use a staggered finite difference approach with cell-centered hydraulic conductivities estimated by averaging nodal conductivities. We evaluate the impact of several averaging schemes (arithmetic, harmonic, geometric and log-mean) on the occurrence of multiple solutions and associated numerical issues. The resulting numerical schemes are compared in terms of their propensity to trigger multiple roots at soil interfaces. Our results show that the choice of averaging scheme does affect the occurrence of multiple solutions and long term behavior of the numerical solution. In particular, our simulations confirm that the averaging schemes associated with larger interface conductivities (log-mean and arithmetic mean) are less likely to suffer from non-uniqueness issues of the interface problem.

INTRODUCTION

The vertical infiltration flow in unsaturated porous medium is described by the one-dimensional Richards' equation (in mixed form),

$$\frac{\partial \theta}{\partial t} = -\frac{\partial q}{\partial z}, \quad q = K \left(1 - \frac{\partial h}{\partial z} \right), \quad (1)$$

where $\theta = \theta(z, t)$ represents volumetric water content, z is depth (positive downwards), q is infiltration flux, K denotes soil hydraulic conductivity, and h is capillary pressure head. In the

soil domain $0 \leq z \leq L$, this governing equation is augmented with initial/boundary conditions in terms of capillary pressure head h or flux q , i.e.,

$$\begin{cases} h(z, 0) = h_0(z), & 0 < z < L, \\ h(0, t) = h_{\text{top}}, \text{ or } q(0, t) = q_{\text{top}}, & t \geq 0, \\ h(L, t) = h_{\text{bot}}, \text{ or } q(L, t) = q_{\text{bot}}, & t \geq 0. \end{cases} \quad (2)$$

The hydraulic relations between θ, h, K and effective saturation $S = (\theta_s - \theta_r)/(\theta - \theta_r)$ are described by soil-water characteristic curves and permeability functions, where θ_s and θ_r are the saturated and residual volumetric water contents. In this work, we investigate three constitutive models, the Gardner model, the Mualem-van Genuchten model (MvG) and the Fredlund & Xing-Leong & Rahardjo model (FXLR). The problem is non-dimensionalized using L and L/K_s as length and time scales, where K_s is the saturated hydraulic conductivity of a reference layer.

Layers in the soil are characterized by different empirical soil hydraulic parameters. The interface between two soil layers leads to a discontinuity in volumetric water content, while both pressure head and flux are continuous across the soil interface. Historically, simulations of water flow across interfaces have been obtained using head-based nodal averaging, for example the arithmetic average of heads (Kirkland et al., 1992) or the (weighted) average of water contents (Zha et al., 2013), without imposing continuity of flux. Commonly used commercial softwares do not impose continuity of flux across interfaces. For example, HYDRUS smoothly changes the soil parameters across the interface of dissimilar soils (Simunek et al., 1998). Numerical oscillations can be triggered at interfaces using HYDRUS, SVFlux and Vadose/W (Dye, 2008). Hills et al. (1989) first pointed out that the flux must be continuous across the interface, and Romano et al. (1998) first proposed an algorithm to enforce continuity of both flux and pressure head numerically at the soil interface using central approximations of head gradient and flux. Others (Schaudt and Morrill, 2002; Matthews et al., 2004) imposed continuity of flux across the interface by using one-sided (nonlocal) approximations of flux on either side of the interface.

In this work, we do enforce continuity of both flux and pressure head at the interface without changing the soil parameters in the vicinity of the interface. The approach in this work is similar to ideas found in Romano et al. (1998). However, Romano et al. (1998) do not address the well-posedness of the interface equation. Liu et al. (2016) show that multiple solutions of the nonlinear interface equation may exist in certain soil parameter regimes, especially when sharp wetting fronts pass through the interface between highly dissimilar materials. A full mathematical analysis of the interface problem in terms of two parameters depending on local hydraulic conditions and mesh size is possible for the quasi-linear Gardner model using the geometric mean for cell-centered hydraulic conductivities (Liu et al., 2016). Non-uniqueness of solutions of the interface equation may also be triggered numerically for more advanced models under certain hydraulic conditions.

The estimation of cell-centered hydraulic conductivities based on nodal conductivities affects the existence of multiple solutions of the interface equation. Four commonly used averaging methods are investigated in this work,

1. harmonic mean

$$K_{j+\frac{1}{2}}^{(h)} = \frac{2K_j K_{j+1}}{K_j + K_{j+1}}, \quad (3)$$

2. geometric mean

$$K_{j+\frac{1}{2}}^{(g)} = \sqrt{K_j K_{j+1}}, \quad (4)$$

3. log mean

$$K_{j+\frac{1}{2}}^{(l)} = \begin{cases} \frac{K_{j+1} - K_j}{\ln(K_{j+1}/K_j)}, & \text{if } K_j \neq K_{j+1}, \\ K_j, & \text{if } K_j = K_{j+1}, \end{cases} \quad (5)$$

4. arithmetic mean

$$K_{j+\frac{1}{2}}^{(a)} = \frac{1}{2}(K_j + K_{j+1}), \quad (6)$$

where K_j and K_{j+1} are the conductivity values at the (nodal) grid points in the finite different scheme, and $K_{j+\frac{1}{2}}$ is a cell-centered conductivity (Figure 1).

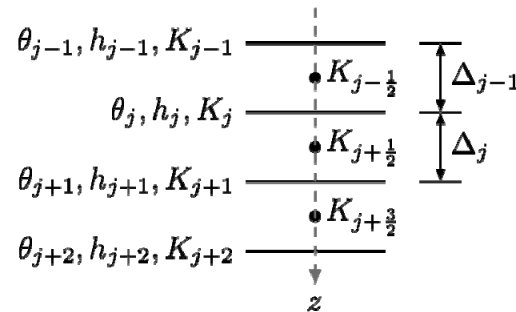


Figure 1. Staggered finite different scheme.

Other averaging methods based on the average of pressure heads substituted into the permeability function $K(h)$ are also commonly used (Haverkamp and Vauclin, 1979; Baker, 2006). However, due to the nonlinearity of permeability functions for unsaturated soils, using pressure head averages to derive conductivity averages can lead to large errors. Moreover, in a layered soil, the permeability function $K(h)$ is unknown at the interface.

MATHEMATICAL FORMULATION

The Interface Problem

Assume the soil column is partitioned into distinct layers with different hydraulic properties, described by different soil hydraulic parameters. Near one interface between two layers, the parameters are superscripted by $^-$ (upstream) and $^+$ (downstream) as shown in Figure 2.

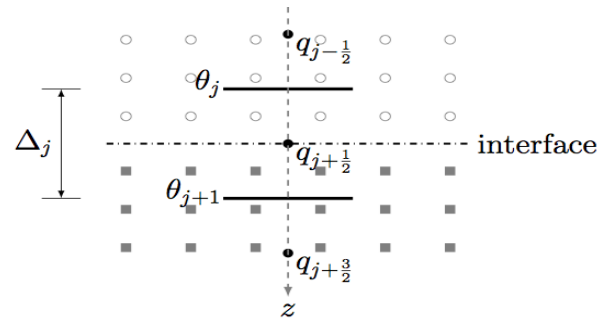


Figure 2. Interface of two distinct soil layers.

Approximating spatial derivatives via standard second-order central finite differences yields a system of N ordinary differential equations (ODEs),

$$\frac{d\theta_j}{dt} = -\frac{q_{j+\frac{1}{2}} - q_{j-\frac{1}{2}}}{\frac{1}{2}(\Delta_{j-1} + \Delta_j)}, \quad (7)$$

with

$$q_{j+\frac{1}{2}} = K_{j+\frac{1}{2}} \left(1 - \frac{h_{j+1} - h_j}{\Delta_j} \right). \quad (8)$$

Across the soil interface, Eq. (7) remains valid because it represents mass conservation through each homogeneous layer. On the other hand, Eq. (8), which involves quantities across the interface, no longer holds. Also, the conductivity at the interface, $K_{j+\frac{1}{2}}$, cannot be estimated by averaging K_j and K_{j+1} , since these quantities are associated with two distinct soil layers. Instead, continuity of flux q at the interface yields an equation

$$q_{j+\frac{1}{2}} := K_{j+\frac{1}{2}}^- \left(1 - \frac{h_{j+1}^- - h_j}{\Delta} \right) = K_{j+\frac{1}{2}}^+ \left(1 - \frac{h_{j+1} - h_j^+}{\Delta} \right). \quad (9)$$

The quantities h_{j+1}^- , h_j^+ represents pressure heads at “ghost” (or fictitious) nodes (Romano et al., 1998) in extended layers (Figure 3). Following Romano et al. (1998) and Liu et al. (2016), these values are defined via linear extrapolation,

$$\begin{aligned} h_{j+1}^- &= 2h_{j+\frac{1}{2}} - h_j = h_{j+1} - \delta h, \\ h_j^+ &= 2h_{j+\frac{1}{2}} - h_{j+1} = h_j - \delta h, \end{aligned} \quad (10)$$

with

$$\delta h := h_{j+1} + h_j - 2h_{j+\frac{1}{2}},$$

taking into account continuity of pressure head h_j across the interface.

The cell-centered hydraulic conductivity $K_{j+\frac{1}{2}}^-$, from the upper layer point of view, is an average of K_j and K_{j+1}^- . Similarly, $K_{j+\frac{1}{2}}^+$, from the lower layer point of view, is an average of K_j^+ and K_{j+1} . The extrapolated conductivities K_{j+1}^- and K_j^+ are evaluated from the “ghost” pressure heads h_{j+1}^- and h_j^+ using the soil parameters in the permeability function $K(h)$ associated with the upper/lower layer. Substituting Eq. (10) into Eq. (9) yields

$$\delta h = \left(\Delta - (h_{j+1} - h_j) \right) \frac{1-r}{1+r} \quad (11)$$

with

$$r := \frac{K_{j+\frac{1}{2}}^-}{K_{j+\frac{1}{2}}^+} = g(r). \quad (12)$$

Eq. (12) with δh given by Eq. (11) represents a nonlinear equation in r , which must be solved iteratively, using either a fixed-point (Picard) iteration or a Newton iteration.

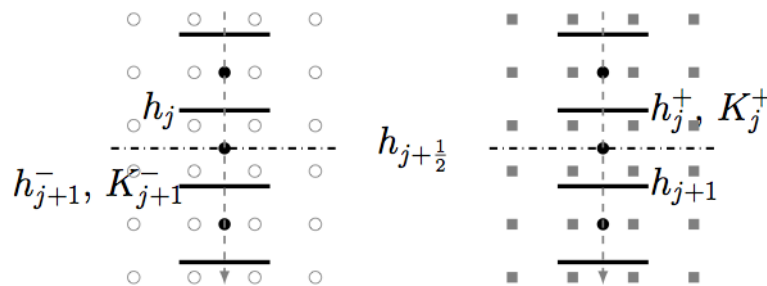


Figure 3. “Ghost” nodes in extended layers. Quantities associated with the upper (lower) layer are referenced with superscript $^+$ ($^-$).

Averaging Methods for Cell-Centered Conductivities

In a one-dimensional vertical soil column, different average hydraulic conductivities in one cell can be interpreted as placing soil layers in different directions. The arithmetic mean Eq. (6) can be interpreted as aligning the layers parallel to each other horizontally (parallel flow) while the harmonic mean Eq. (3) is interpreted as stacking the layers vertically (serial flow). When the values of two conductivities, K_j and K_{j+1} , are very different, the arithmetic mean tends to yield a large average, as if water flow chooses the faster route among the parallel channels. The harmonic mean results in a smaller average value because the lower conductivity behaves like a bottleneck to water flow. Physically, the harmonic mean approximates one-dimensional water flow in a layered soil column more precisely than the arithmetic mean. However, using the harmonic mean also leads to more numerical difficulties.

The geometric mean Eq. (4) is based on the arithmetic mean of pressure heads using the Gardner model, i.e., the relative conductivity $K(h) = e^{\alpha h}$. Since the average of pressure heads is $\bar{h} = (h_j + h_{j+1})/2 = (\ln K_j + \ln K_{j+1})/2\alpha$, the average conductivity $e^{\alpha \bar{h}}$ yields the geometric mean of K_j and K_{j+1} . Notice that the geometric mean is based on the assumption that pressure head is linear within a cell. However, when the cell straddles two different soil layers, pressure head is not linear. Using a linear approximation of the pressure head may introduce numerical errors.

The log mean Eq. (5) is also based on the Gardner relation, $K(h) = e^{\alpha h}$, but derived from the average conductivity within one cell,

$$K_{j+\frac{1}{2}} = \frac{1}{h_{j+1} - h_j} \int_{h_j}^{h_{j+1}} K(h) dh = \frac{K_{j+1} - K_j}{\ln(K_{j+1}/K_j)}. \quad (13)$$

It can be easily verified that

$$K_{j+\frac{1}{2}}^{(h)} \leq K_{j+\frac{1}{2}}^{(g)} \leq K_{j+\frac{1}{2}}^{(l)} \leq K_{j+\frac{1}{2}}^{(a)}, \quad (14)$$

with equality for $K_j = K_{j+1}$.

In a layered soil, using different averaging methods for cell-centered conductivities affects the well-posedness of the interface problem Eq. (9). In this work, the robustness of various averaging methods is investigated by performing numerical experiments in multi-layer soils.

NUMERICAL EXPERIMENTS

Three constitutive models are investigated here. We note that the soil hydraulic parameters are not directly convertible between different hydraulic models, and that numerical issues involving multiple solutions of the interface equation are dependent on the soil hydraulic models.

The numerical setup is shown in Figure 4. The soil consists of several layers. Each layer is associated with a set of soil hydraulic parameters. We use a staggered finite difference method with a uniform spatial discretization for each layer (Figure 4).

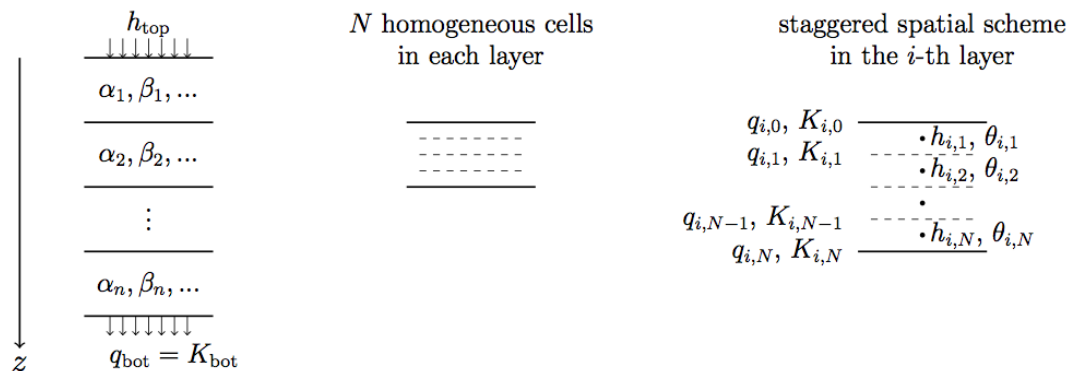


Figure 4. Multi-layer soil.

Gardner Model

The Gardner model (Gardner, 1958) is a simple hydraulic conductivity relation,

$$S = \begin{cases} e^{\alpha h} & \text{if } h < 0, \\ 1 & \text{if } h \geq 0, \end{cases} \quad K = \beta S, \quad (15)$$

with $\alpha > 0, \beta > 0$. Although the Gardner model is rarely used in practice, it is often used in numerical experiments as a reference to describe soil-water characteristics in many situations, and as a test problem for investigative purposes.

Numerical experiment is performed in a multi-layer soil with five alternating layers with uniform thickness. Each layer is discretized into 20 uniform spatial cells. The odd layers (layer 1, 3 and 5) are assigned non-dimensionalized soil parameters $\alpha = 13, \beta = 1$, and the even layers

(layer 2 and 4) with $\alpha = 1, \beta = 0.0006$. The residual and saturated water contents are the same for all layers, $\theta_r = 0.06, \theta_s = 0.4$. The initial pressure head profile is $h(z, 0) = -1, 0 < z < 1$. The boundary conditions are $h(0, t) = -0.4$ (fixed pressure head) and $q(1, t) = K(1, t)$ (free drainage) for $t \geq 0$. Figure 5 shows the numerical solutions of pressure head profile at different times using the four averaging methods for cell-centered conductivities. The occurrence of multiple solutions to the interface problems is triggered across the interfaces at $z = 0.2$ and 0.6 , when using the harmonic mean and geometric mean. At $t = 1$, there is numerical ponding near the interface $z = 0.2$ for the harmonic and geometric means. For the geometric mean, the multiple solutions to the interface problem disappear shortly after the wetting front has passed through this interface, then the numerical result autocorrects itself at $t = 3$ due to the diffusive nature of the flow (and Richards' equation). However, the multiple solutions subsist after they first occur for the harmonic mean, and an incorrect infiltration profile persists because of low conductivities at the interfaces from a high conductivity layer to a low conductivity layer. Similarly behavior is shown for the harmonic mean and geometric mean due to the existence of multiple solutions across the interface $z = 0.6$ at $t = 10$ and 100 . The log-mean and arithmetic mean in this case always admit a unique solution to the interface problems.

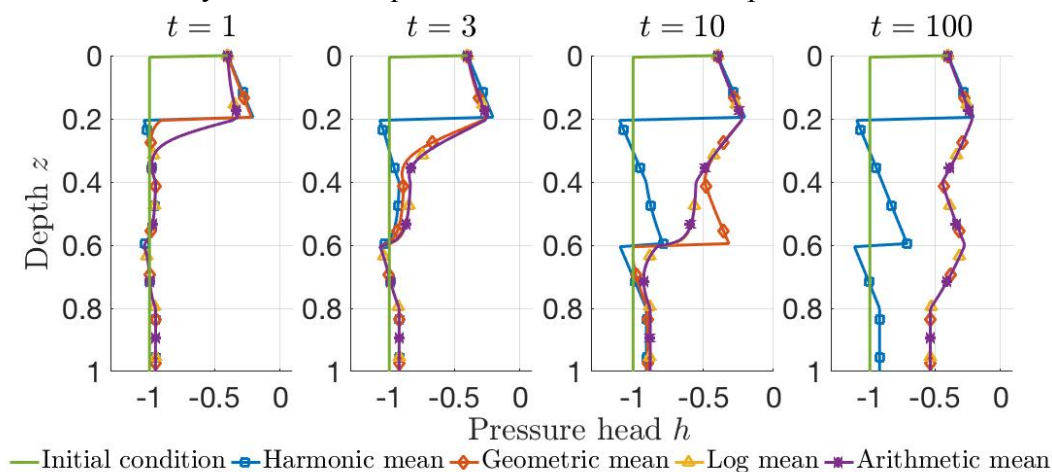


Figure 5. Numerical simulations at different times of a 1D infiltration process in a five-layer soil using the Gardner model. Soil interfaces are $z = 0.2, 0.4, 0.6$ and 0.8 .

MvG Model

The Mualem-van Genuchten (MvG) model (van Genuchten 1980) is

$$S = \begin{cases} (1 + (-ah)^n)^{-m} & \text{if } h < 0, \\ 1 & \text{if } h \geq 0, \end{cases} \quad (16)$$

$$K = \beta \sqrt{S} \left(1 - \left(1 - S^{\frac{1}{m}} \right)^m \right)^2,$$

with $\alpha > 0, \beta > 0, n > 1$, and $0 < m = 1 - 1/n < 1$.

Numerical simulation for the MvG model in a ten-layer soil is investigated. Each layer with uniform thickness is discretized into 20 uniform spatial cells. The odd layers (1, 3, 5, 7 and 9) are assigned with non-dimensionalized soil parameters $\alpha = 0.13, \beta = 1, n = 2.5$, and the

even layers (2, 4, 6, 8 and 10) with $\alpha = 0.29, \beta = 1, n = 2.72$. The residual and saturated water contents are $\theta_r = 0.001, \theta_s = 0.368$ for all the layers. The boundary conditions are fixed pressure head values for the top and the bottom of the soil, $h(0, t) = -1$ and $h(1, t) = -135$ for $t \geq 0$. The initial pressure head profile is $h(z, 0) = -136 + z$ for $0 < z < 1$. Figure 6 shows the space-time relation representing where and when a given pressure head, $h = \ln((e^{h(0,t)} + e^{h(1,t)})/2) \approx -1.693147$, is obtained.

When using the harmonic mean, the cell-centered conductivities are extremely underestimated so the numerical water infiltration is too slow to be acceptable. The wetting front has not reached the top-most interface at $z = 0.1$ when $t = 0.1$. The other three averaging methods yield more acceptable results. However, using the geometric mean triggers multiple solutions to the interface problems, leading to oscillations in the curve in Figure 6. In this experiment, the log-mean and arithmetic mean always admit a unique solution to the interface problems.

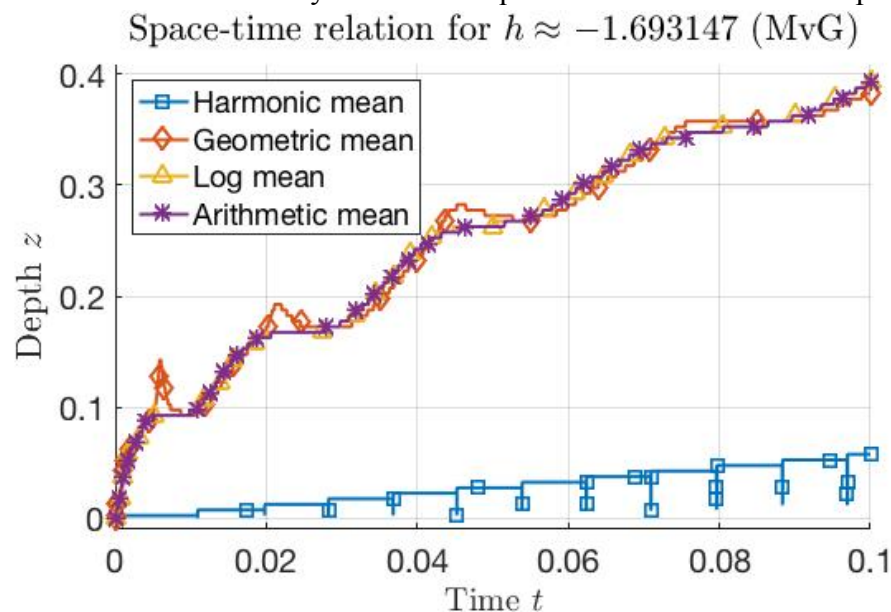


Figure 6. Depth v.s. time relation for where and when $h \approx -1.693147$ is reached in a 1D infiltration process in a ten-layer soil setting using the MvG model.

FXLR Model

The Fredlund & Xing-Leong & Rahardjo (FXLR) model (Fredlund and Xing 1994; Leong and Rahardjo 1997) is

$$S = \begin{cases} [\ln(e + (-\alpha h)^n)]^{-m} & \text{if } h < 0, \\ 1 & \text{if } h \geq 0, \end{cases} \quad (17)$$

$$K = \beta S^p,$$

with $\alpha > 0, \beta > 0, n > 0, m > 0$, and $p > 0$.

We use a soil column with 10 uniform layers discretized into 20 uniform cells each. The soil hydraulic parameters are $\alpha = 0.015, n = 2.5, m = 5, p = 18$ for all the layers. The odd layers are assigned with $\beta = 1$ and the even layers $\beta = 0.9$. The initial/boundary conditions are $h(z, 0) = -70$ for $0 < z < 1$, and $h(0, t) = -40, h(1, t) = -70$ for $t \geq 0$.

Figure 7 shows the space-time relation for obtaining a given pressure head ($h = \ln((e^{h(0,t)} + e^{h(1,t)})/2) \approx -40.693147$). The numerical infiltration using the harmonic mean is too slow as shown in Figure 7, similar to the numerical results of the MvG model. Oscillations are observed for the geometric mean due to the existence of multiple solutions to the interface problems. Numerical ponding occurs at even soil interfaces despite the relatively small dissimilarity between the saturated conductivities of even and odd layers, due to ill-posedness of the interface problems when using the geometric mean. We notice that multiple solutions still exist in a homogeneous soil column (the same β value for all the layers), but the oscillations (between $t \approx 50$ and 450) in Figure 7 for the geometric mean disappear because $r = 1$ is always obtained by the iteration method for Eq. (12). In this experiment, the log-mean and arithmetic mean again admit a unique solution and exhibit similar behaviors of the infiltration process.

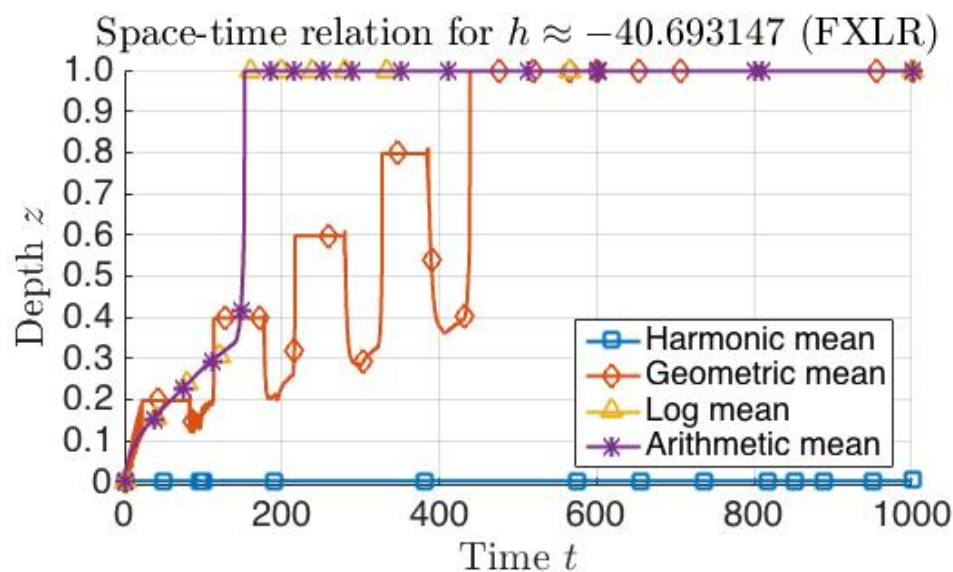


Figure 7. Depth v.s. time relation for where and when $h \approx -40.693147$ is reached in a 1D infiltration process in a ten-layer soil setting using the FXLR model.

CONCLUSION

The results in this study show that the averaging scheme used for cell-centered conductivities impacts the ill-posedness of the interface problem. The existence of multiple solutions to the interface problem can lead to a non-physical infiltration process such as numerical ponding or numerical oscillations. In a layered soil, using a relatively large average conductivity (for example, as obtained from the log-mean and arithmetic mean) is less likely to trigger multiple solutions to the interface problem.

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REFERENCES

- Baker, D. L. (2006). "General validity of conductivity means in unsaturated flow models." *Journal of Hydrologic Engineering*, 11(6), 526-538.
- Dye, H. B. (2008). "Moisture movement through expansive soil and impact on performance of residential structures." Ph.D. thesis, Arizona State University, Tempe, Arizona, USA. (May).
- Fredlund, D. G. and Xing, A. (1994). "Equations for the soil-water characteristic curve." *Canadian geotechnical journal*, 31(4), 521-532.
- Gardner, W. (1958). "Some steady-state solutions of the unsaturated moisture flow equation with application to evaporation from a water table." *Soil science*, 85(4), 228-232.
- Haverkamp, R. and Vauclin, M. (1979). "A note on estimating finite difference interblock hydraulic conductivity values for transient unsaturated flow problems." *Water Resources Research*, 15(1), 181-187.
- Hills, R., Porro, I., Hudson, D., and Wierenga, P. (1989). "Modeling one-dimensional infiltration into very dry soils: 1. Model development and evaluation." *Water Resources Research*, 25(6), 1259-1269.
- Kirkland, M. R., Hills, R., and Wierenga, P. (1992). "Algorithms for solving Richards' equation for variably saturated soils." *Water Resources Research*, 28(8), 2049-2058.
- Leong, E. C. and Rahardjo, H. (1997). "Permeability functions for unsaturated soils." *Journal of Geotechnical and Geoenvironmental Engineering*, 123(12), 1118-1126.
- Liu, R., Welfert, B., and Houston, S. (2016). "Numerical Issues Arising in Determination of Interlayer Conductivities in Layered Unsaturated Soils." *International Journal of Geomechanics*, 04016078.
- Matthews, C. J., Braddock, R. D., and Sander, G. (2004). "Modeling flow through a one-dimensional multi-layered soil profile using the Method of Lines." *Environmental Modeling & Assessment*, 9(2), 103-113.
- Romano, N., Brunone, B., and Santini, A. (1998). "Numerical analysis of one-dimensional unsaturated flow in layered soils." *Advances in Water Resources*, 21(4), 315-324.
- Schaudt, K. and Morrill, J. (2002). "On the treatment of heterogeneous interfaces in soil." *Geophysical research letters*, 29(10), 34-1.
- Simunek, J., Huang, K., and van Genuchten, M. T. (1998). "The HYDRUS code for simulating the one-dimensional movement of water, heat, and multiple solutes in variably-saturated media." *US Salin. Lab. Res. Rep.*, 144.
- van Genuchten, M. T. (1980). "A closed-form equation for predicting the hydraulic conductivity of unsaturated soils." *Soil science society of America journal*, 44(5), 892-898.
- Zha, Y., Yang, J., Shi, L., and Song, X. (2013). "Simulating one-dimensional unsaturated flow in heterogeneous soils with water content-based Richards equation." *Vadose Zone Journal*, 12(2).